



CANDIDATE
NAME

CLASS

2T

INDEX
NUMBER

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BIOLOGY
Paper 2: Core Paper

9648/02
26th August 2014
2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are two sections in this paper.

Section A]

Answer all questions

Section B]

Answer **ONE** question. Answer each part on a **separate** piece of paper.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

For Examiner's Use	
Section A	80
1 [10]	
2 [10]	
3 [10]	
4 [10]	
5 [10]	
6 [10]	
7 [10]	
8 [10]	
Section B	20
7a / 8a [7]	
7b / 8b [7]	
7c / 8c [6]	
TOTAL	100

Section A

Answer **all** questions in this section.

- 1 All cells are surrounded by a cell surface membrane. There are also membrane systems within the cell, as well as several types of membrane-bound organelles. Some of these have a single membrane surrounding them, while others have two. All cell membranes have a similar basic structure but the proportions of the various molecules present vary between membranes and even between the inner and outer layers of a membrane.

Fig. 1.1 shows a generalised cell surface membrane.

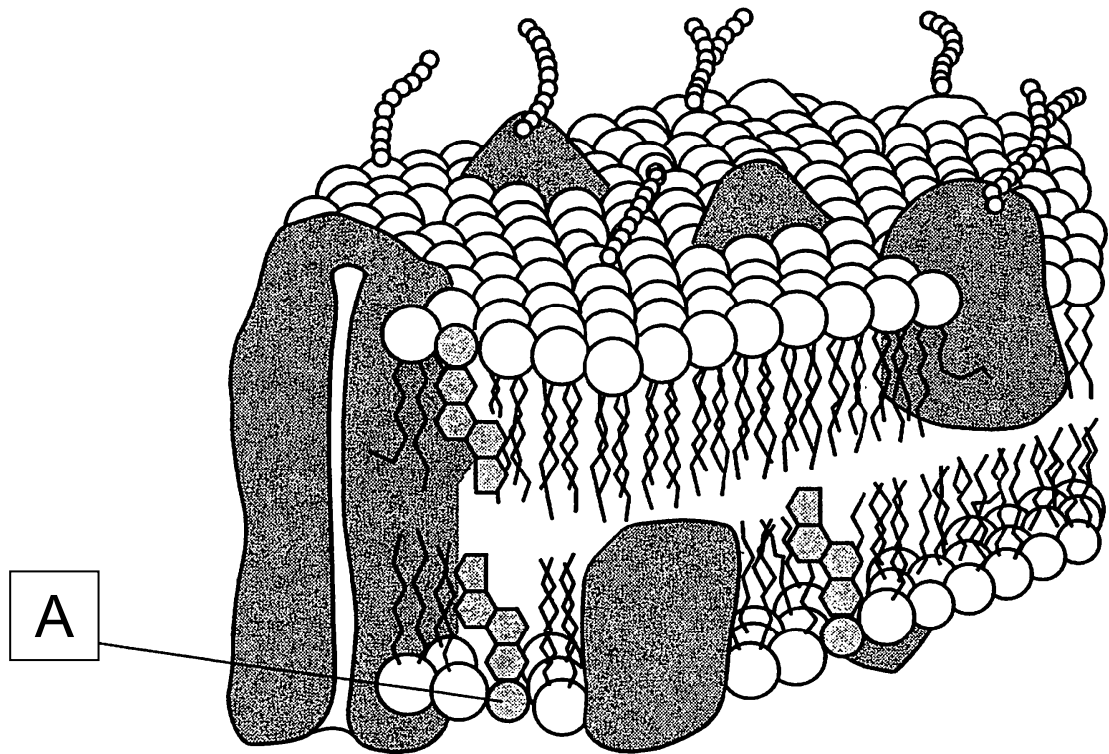


Fig. 1.1

With reference to Fig. 1.1,

- (a) (i) identify the molecule A and describe its function.

[2]

- Cholesterol
- regulates/maintain membrane fluidity

- (ii) With reference to Fig. 1.1, state one way in which the outer surface membrane is different from the inner surface.

[1]

1. presence of glycoprotein and glycolipid in the outer membrane where none occur on the inner surface membrane.

- (iii) state one functional property resulting from this difference

[1]

1. outer membrane able to adhere / attach with other cells with complementary glycoprotein and / or glycolipid matrixes
2. the potential for cell to cell recognition
3. cell signaling

(b) Table 1.1 shows the approximate percentage of protein in different membranes.

Table 1.1		
Organelle	Membrane	Protein as a percentage of dry weight.
chloroplast	outer membrane	54
chloroplast	inner membrane	70
mitochondrion	outer membrane	55
mitochondrion	inner membrane	80

(b)(i) Using data from Table 1.1, describe **one similarity** in the distribution of proteins within the two membranes of chloroplast and mitochondrion and suggest a reason. [3]

Similarity	1. Inner membranes of both mitochondria and chloroplast are higher in protein percentage compared to their respective outer membranes. Mitochondria (inner 70 vs outer 54) / Chloroplast (inner 80 vs outer 55) 2. Outer membranes protein percentage in both mitochondria and chloroplast were almost equal (mitochondria 55 vs Chloroplast 54) Reason x1 mark Data cited x1mark
Reason	1. Inner membrane protein percentage is higher due to involvement of proteins with intra cellular matrix and enzymatic roles, compared to mainly transport proteins in the outer membrane. 2. the outer membrane protein percentage was almost identical due to similar roles of mainly transport Max 1

(b) (ii) Describe and explain one physical way in which the inner membrane of a mitochondrion is adapted to its function. [3]

1. many folds (cristae) of the inner mitochondrial membrane.
2. presence of electron transport chain and ATP synthase for ATP production.
3. increase surface area- allows the more elements of ETC and Stalked particles to be grouped together in close proximity to one another.
4. membrane is impermeable to hydrophilic molecules- allowing a build up / reservoir of hydrophilic ions that are later involved in chemiosmosis.

[Total: 10]

- 2 Protein kinases are a class of enzymes that are involved in the triggering of the cell cycle. Kinases add phosphate to molecules, and the modification can serve as a "switch" to turn events in the cell on or off. Cdk or cyclin dependent kinases (Cdk) regulate the cell cycle. Fig. 2.1 shows a representation of Cdk

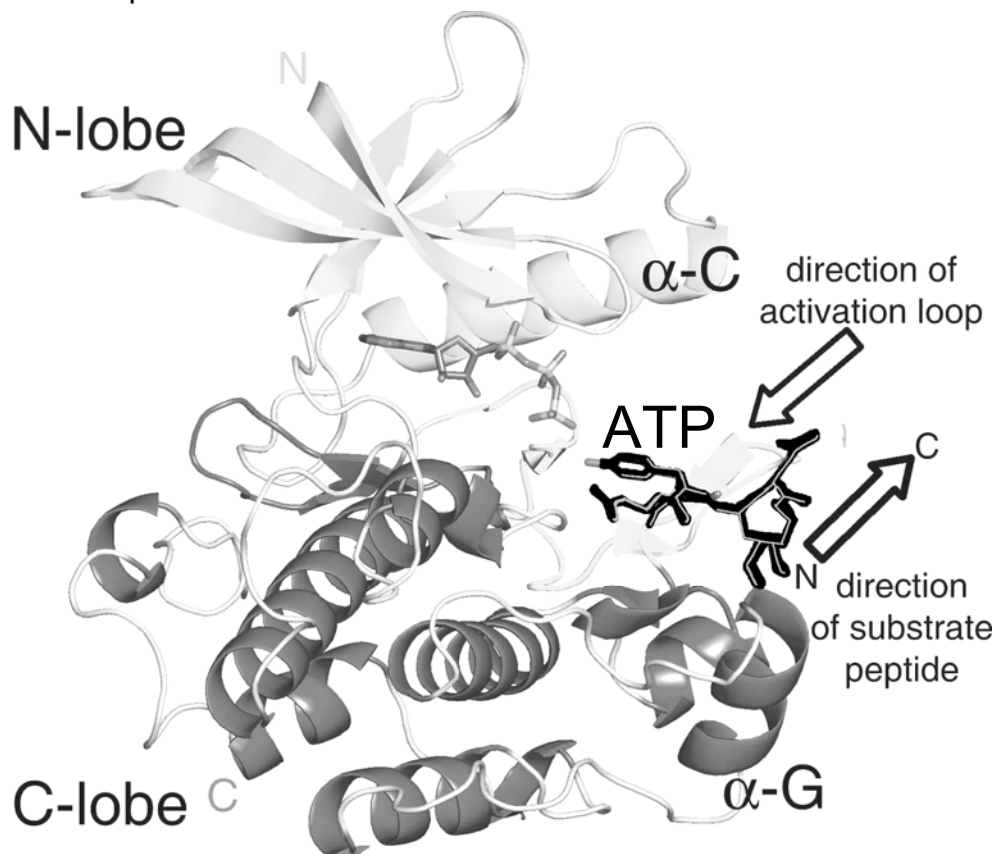


Fig. 2.1 Anatomy of a Cdk catalytic domain. The substrate peptide, docked onto the activation loop, and an ATP analog are shown in black. α -Helix-C, which contains residues critical for catalysis, and α -helix-G, which is involved in docking protein substrates, are labeled.

- (a) Describe how specialised enzymes are structurally specific to their substrate [2]
1. enzymes are globular in shape and are in specific configuration that is complementary to the substrate conformation
 2. Lock and Key hypothesis.

With reference to Fig. 2.1,

- (b) explain the specific significance of the following parts of protein kinase [2]

Feature	Significance
N-Lobe and C-Lobe	1] maintains specific globular configuration of protein kinase
α -C and α -G	Catalytic R Groups and binding of the R groups
directions of activation loop and substrate peptide	2] once catalytic R groups attach to substrate causes a strain in substrate due to opposing directions of the activation loop and substrate peptide. this lowers activation energy of the reaction

Cyclin-dependent kinases (Cdk) are a family of protein kinases first discovered for their role in regulating the cell cycle.

(c) Explain how Cdk may be regulated in its action

[1]

1. Cdk are enzymes therefore regulation of protein kinase via competitive inhibition
2. regulation of protein kinase via restriction of substrate.
3. Cdk active only when complexed with cyclins
4. use of phosphatases
5. ATP as a control of substrate

Cdk works always in conjunction with cyclins (another cofactor in the regulation of the Cell cycle) the complexes that are formed when these two are combined makes Cdk active.

Fig. 2.2 shows the relative levels of active Cdk (protein kinase activity) during the cell cycle.

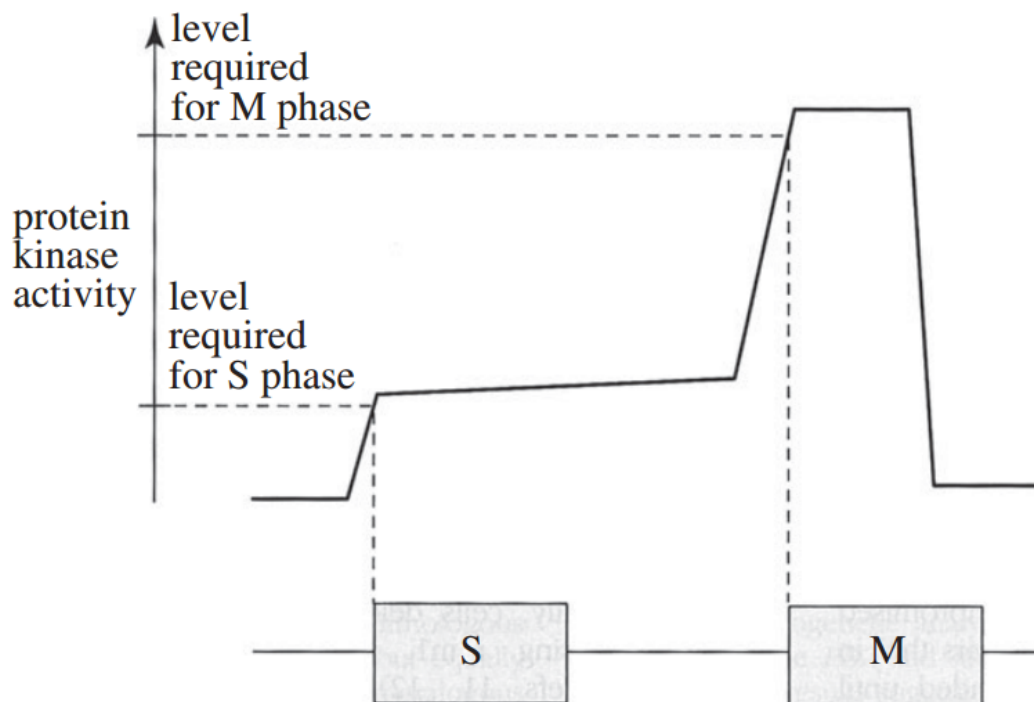


Fig 2.2

(d) Explain what takes place in the gap in-between S phase and M phase (cell division). [3]

1. The gap represents the G2 phase in cell cycle where rapid cell growth takes place
2. in G2 there is the checking on the process of replication and repair
3. in G2 if replication is successful then it passes the G2 checkpoint which is the control point in cell division.

(e) Explain how Cdk is used in the regulation of the cell cycle.

[2]

1. CDKs control the initiation of the cell cycle, they are precursors / activators to each stage of the cell cycle and must be present in critical amounts as indicated by the threshold limits for S phase and a higher threshold limit for Mitotic / meiotic / cell division (OWTTE)
2. Presence of Cdk above respective threshold levels are required for both S and M phases of the cell cycle
3. to allow time for the intermediate processes between S phase and M phase to complete.

[Total: 10]

3 Fig. 3.1 shows the elongation phase of the cellular process of transcription.

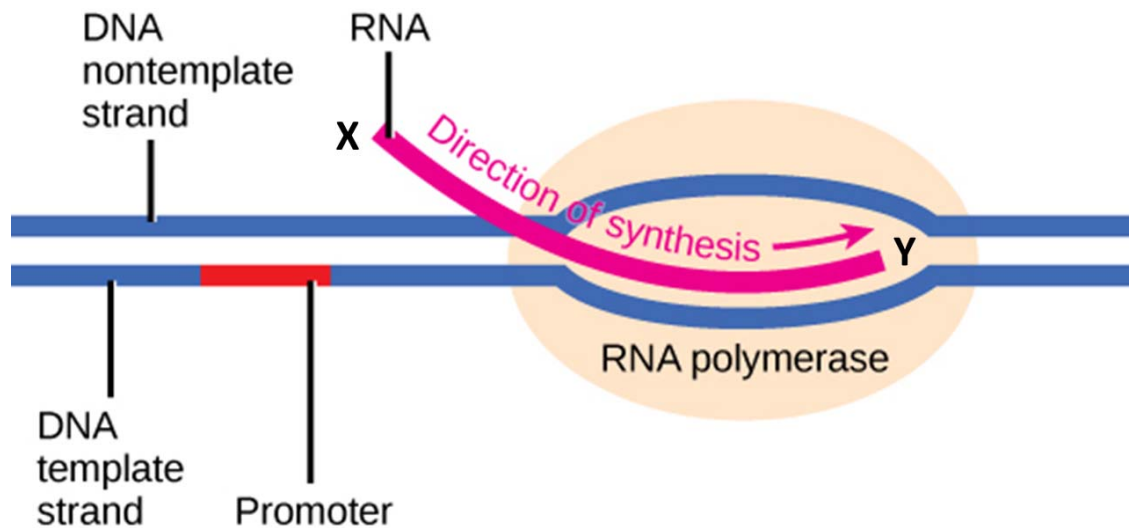


Fig. 3.1

(a)(i) Identify the directionality of the RNA transcript at ends X & Y. [1]

X: 5'

Y: 3'

(ii) State what are the substrates which are not shown in Fig. 3.1 that RNA polymerase allows into its active site for the process of elongation to take place. [1]

- Ribonucleoside triphosphates

(iii) The elongation of RNA transcripts requires energy. Explain how this energy is derived. [1]

1. The energy to drive this process is obtained from the hydrolysis of the incoming ribonucleoside triphosphate molecules, which releases a pyrophosphate.

Fig. 3.2 shows that in breast cancer cells, there is an overexpression of human epidermal growth factor receptor 2 (HER2) protein.

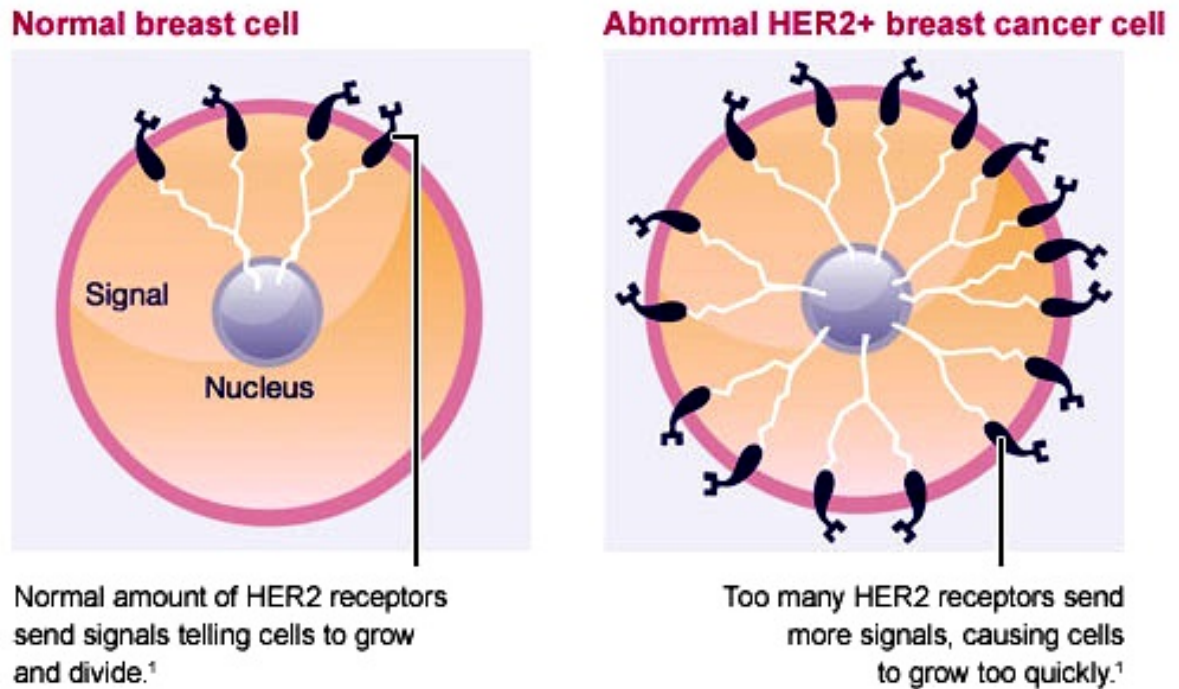


Image reproduced from National Cancer Institute webpage

Fig. 3.2

(b) (i) With reference to Fig. 3.2, suggest why there is an overexpression of HER2 protein. [2]

1. There is an amplification of the HER2 gene.
2. With more HER2 genes present, there is higher rate of HER2 mRNA and protein synthesis/OWTTE.

(ii) Describe and explain another cause not shown in Fig. 3.2 which could result in an overexpression of the HER2 protein. [2]

1. There could have been a mutation in the promoter of the HER2 gene
2. that resulted in constitutive expression of the HER2 gene.

(iii) Suggest with reasons whether HER2 is an oncogene or a tumour suppressor gene. [2]

1. HER2 is an oncogene
2. as there is an overexpression of the same gene / There is a gain of function mutation of the gene.
3. unlike tumour suppressor genes, in which the gene will be inactivated or repressed.

Fig. 3.3 shows the breast cancer drug, Herceptin® which is a monoclonal antibody that binds to HER2 on the cell surface membrane.

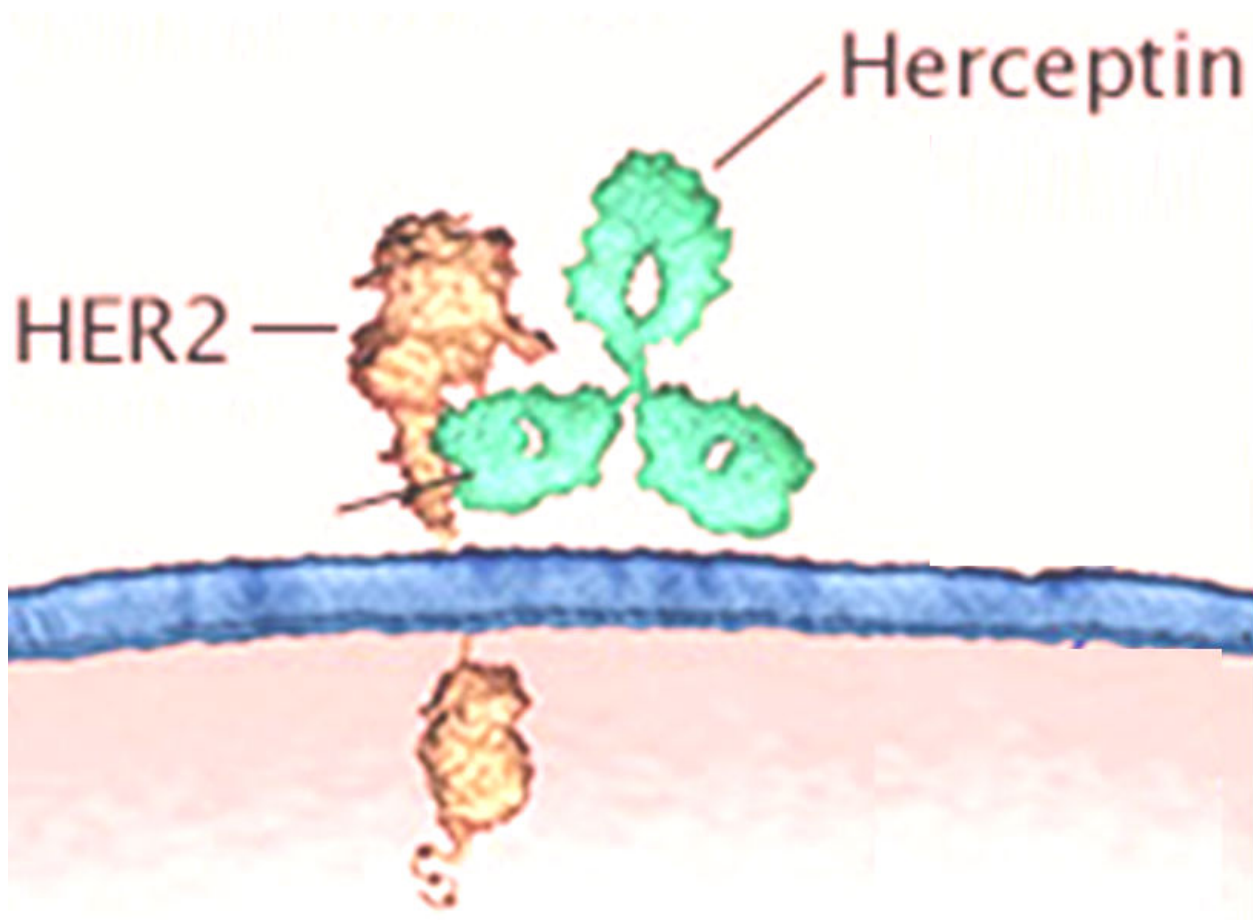


Image modified from <http://www.perjeta.com>

Fig. 3.3

(c) With reference to Fig. 3.3, suggest how Herceptin® stifles the development of cancer. [1]

1. Herceptin binds to the HER2 receptor and thus prevents signals transduction from and thus blocks out cell signaling which will activate pathways leading to cell proliferation.

[Total: 10]

- 4 Fig. 4.1 shows regulation of the arabinose operon. The structural genes of the arabinose operon are *araB*, *araA* & *araD*. These structural genes code for enzymes that are responsible for the generation of xylulose 5-phosphate from arabinose. The initiator sites comprise the operator site (*araO*) and the promoter site (*araI*), whereas the activator protein is encoded by *araC*.

In the presence of arabinose, AraC protein and the CAP-cAMP complex bind to different regions of *araI* resulting in the transcription of its structural genes (Fig. 4.1A). In the absence of arabinose, AraC protein binds to both *araI* & *araO*, which results in a DNA loop that prevents transcription from taking place (Fig. 4.1B).

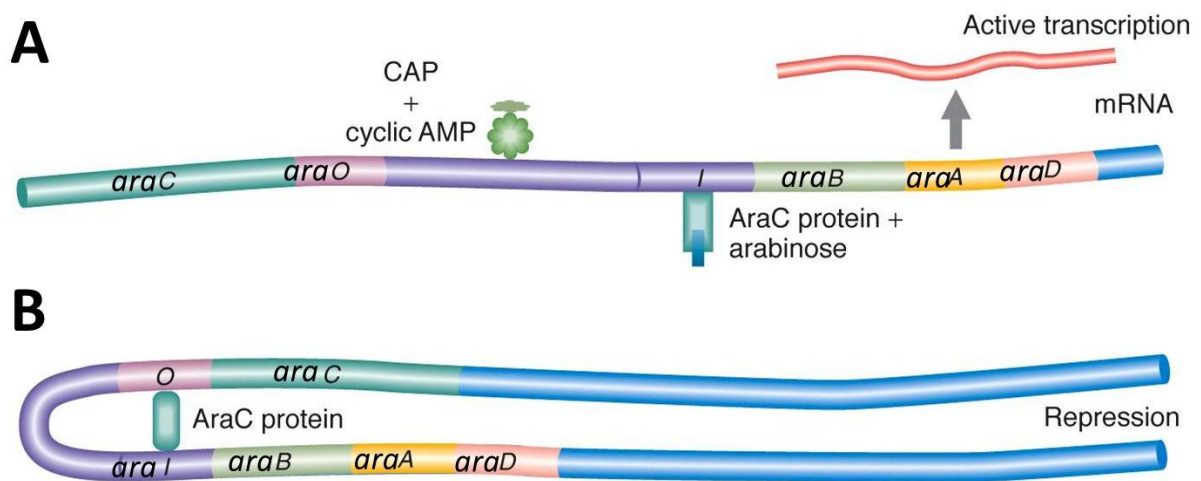


Image modified from Memorial University webpage

Fig. 4.1

- (a) State and explain which type of metabolic reaction arabinose operon is involved in. [2]
1. Arabinose operon is involved in catabolic reactions
 2. which entails the hydrolysis/breakdown of arabinose to xylulose 5-phosphate.

The arabinose operon is controlled by both positive & negative regulation.

With reference to Fig. 4.1, explain how the arabinose operon exhibits:

- (b) (i) positive regulation [2]
1. Only in the presence of arabinose, will arabinose will bind to AraC, preventing the activator protein from binding to *araO*, thus allowing RNA polymerase to bind to the *araI*.
 2. In addition, the CAP-cAMP complex is also needed to bind to *araI* to improve the affinity of RNA polymerase to the promoter.
- (ii) negative regulation [2]
1. In the absence of arabinose, AraC will be able to bind *araO* in addition to *araI*, thus forming a DNA hairpin loop.
 2. RNA polymerase will be unable to bind to the promoter for transcription to take place.

(c) A mutation resulted in the arabinose operon system to be permanently repressed. Suggest where and what type of mutation could have occurred.

[1]

1. The mutation may have occurred in *araC* gene which caused the AraC protein to bind constitutively to *araI*

OR

2. The mutation may have occurred in *araC* gene which caused the binding site on AraC protein for arabinose to change and hence, arabinose will be unable to bind to it.

The Middle East Respiratory Syndrome (MERS) is a viral respiratory disease that is caused by a coronavirus known as MERS-CoV. In 2012, the first case of MERS was reported in Saudi Arabia. Thereafter, other countries near and within the Arabian Peninsula also reported confirmed cases of MERS. Moreover, infected individuals from the affected Middle Eastern countries spread the virus to other countries when they travelled overseas. Individuals, who were infected with MERS-CoV, usually go on to develop severe acute respiratory illness. To date, more than 100 people have died from MERS.

Fig. 4.2 shows the life cycle of MERS-CoV.

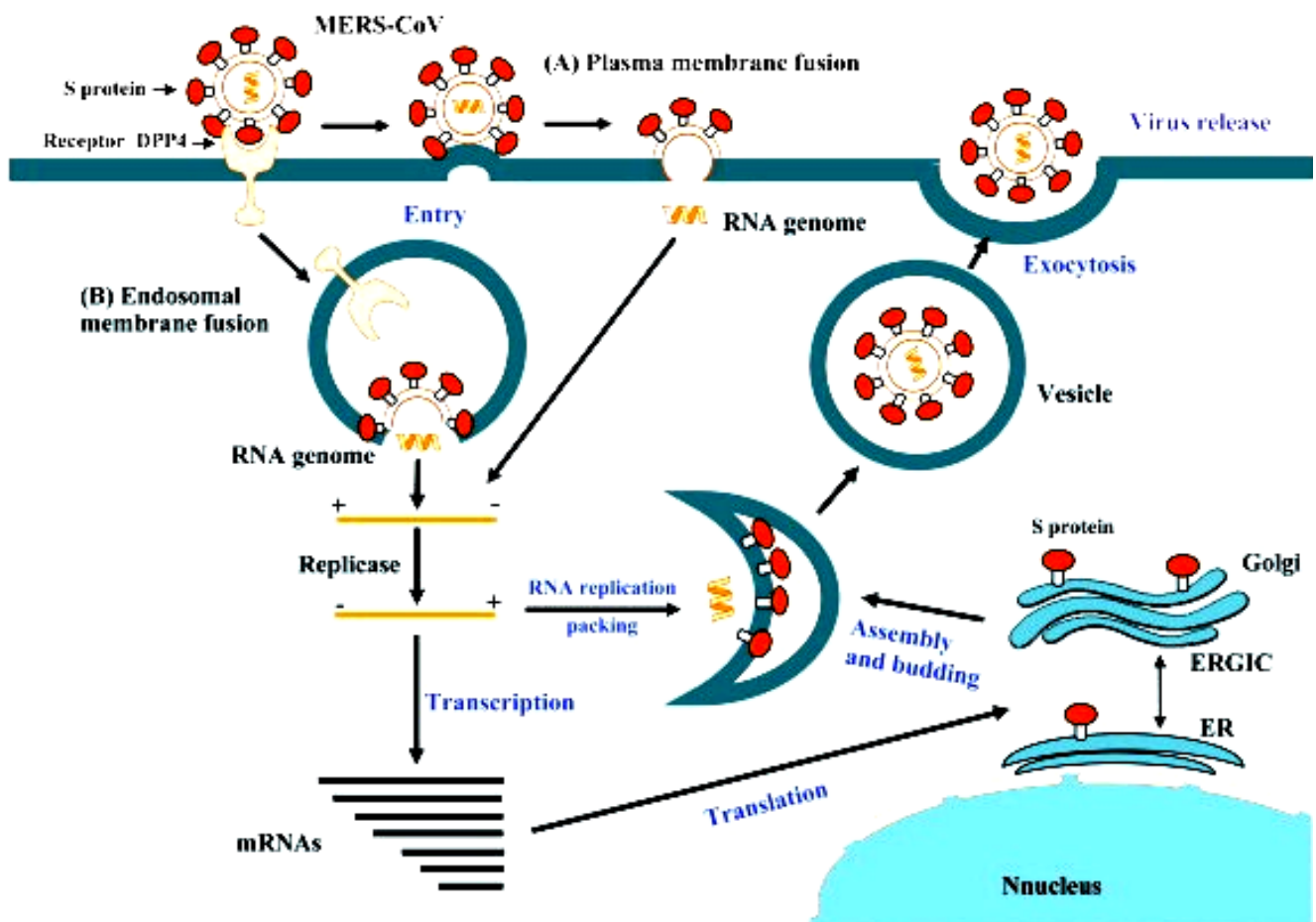


Image reproduced from the article: Middle East respiratory syndrome coronavirus (MERS-CoV): challenges in identifying its source and controlling its spread.

Fig. 4.2

(d) Assuming you are a researcher interested in designing drug to treat MERS, suggest with reference to Fig. 4.2 how you would go about inhibiting the MERS-CoV life cycle. [2]

1. The drug could be a competitive inhibitor that binds to the DPP4 receptor on the host cell surface membrane.
2. This will prevent the host cell from recognizing the virus and hence will not allow it to enter the cell.
3. Inhibitors could be designed to bind to and inhibit the S-protein on MERS-CoV.
4. As S-protein appears to be an important viral surface glycoprotein that mediates the attachment of viruses to the host cell receptor, viral recognition and hence entry will be impaired.
5. Drugs can be designed to inhibit viral replicase.
6. With replicase enzyme inhibited, positive-strand RNA cannot be synthesized using the negative-stranded RNA genome as a template. Hence, viral translation to produce viral proteins cannot take place / RNA replication cannot take place.

Camels in Qatar, Oman, Egypt and Saudi Arabia were found to be also infected with MERS-CoV and it has been hypothesised that MERS-CoV spreaded to humans who came into close contact with infected camels.

(e) Suggest how MERS-CoV evolved to infect both humans and camels. [1]

1. The glycoprotein, gene coding for the S-protein on MERS-CoV could have mutated causing a conformational change that allows its binding on receptors found on humans cells in addition to the receptors found in camel cells.

[Total: 10]

5 Fig. 5.1 shows the binding of a general transcription factor to part of the promoter.

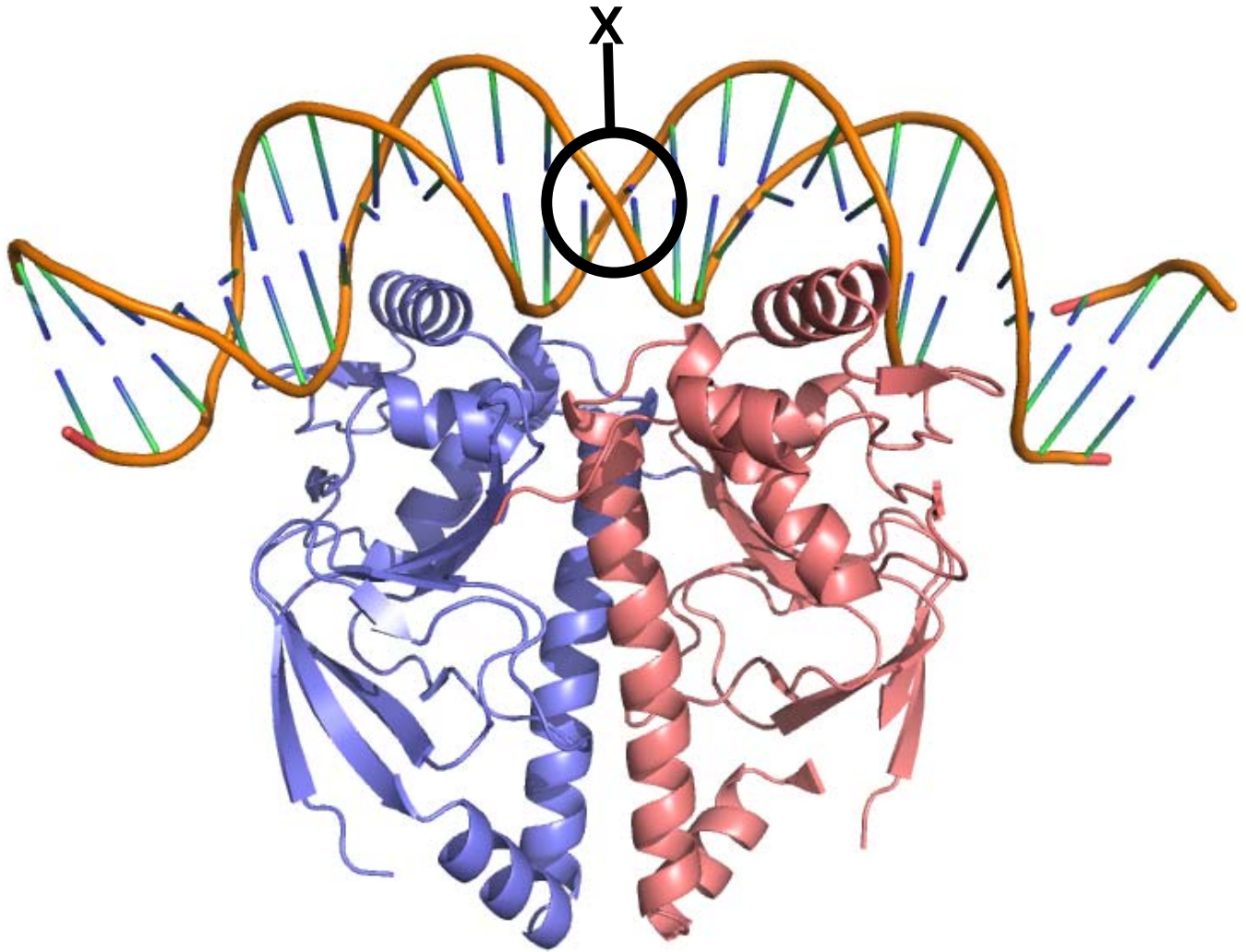


Fig. 5.1

(a) State the function of the general transcription factor.

[1]

- It recruits / facilitates the binding of RNA polymerase to the promoter.

(b) Explain how a substitution mutation occurring at X (circled region) would affect gene expression. [2]

1. The mutation causes the promoter sequence to be no longer complementary to the binding site of transcription factors / transcription factors unable to bind promoter
2. RNA polymerase cannot be recruited to the promoter / transcription cannot be initiated
3. and hence transcription is not initiated / transcription initiation complex cannot be formed.

(c) Describe how proteins, other than general transcription factors, can increase the rate of transcription of a gene beyond basal level. [2]

1. Activators can bind to distal control elements / enhancers...
2. ...interact with other proteins, causing DNA to bend.
3. hence positioning the active transcription initiation complex on the promoter to increase the rate of transcription.

Fig 5.2 below shows a Philadelphia chromosome which is the result of reciprocal translocation between chromosome 9 and chromosome 22. This chromosomal abnormality is often associated with chronic myeloid leukemia (CML) which is cancer of the white blood cells.

- Abl gene expresses a membrane-associated tyrosine kinase which activates a number of cell cycle control proteins and enzymes.
- It is found that bcr/abl fusion gene expresses a mutant tyrosine kinase that is continuously activated which leads to unregulated cell division.
- Tyrosine kinase activity of bcr/abl fusion protein is elevated relative to wild-type abl protein.
- In addition, cells with bcr/abl fusion protein are resistant to apoptosis induced by DNA damage.

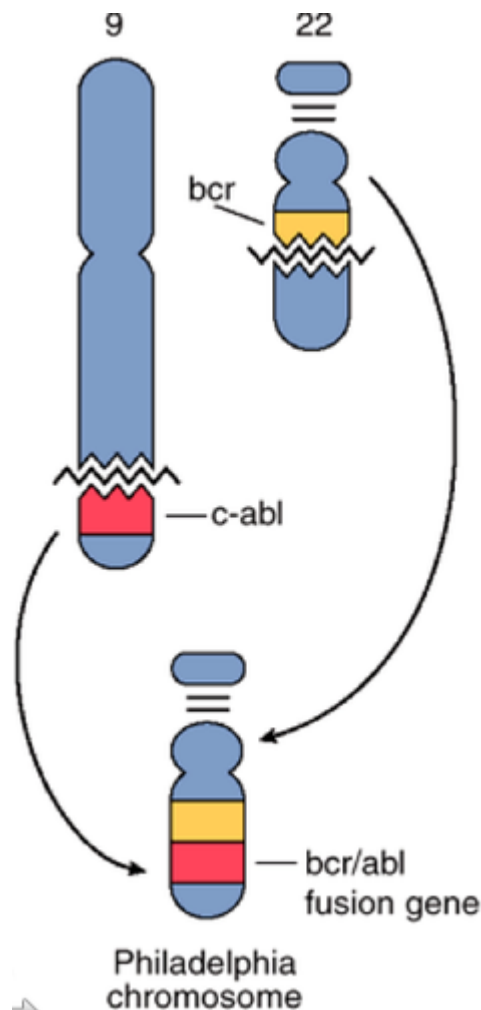


Fig. 5.2

(d) With reference to Fig. 5.2 and the information provided,

(i) state the resultant type of mutation and explain your answer. [2]

1. Gain-of-function mutation. (reject: translocation)
2. The chromosomal translocation resulted in *abl/bcr* fusion protein with increased tyrosine kinase activity / continually activated.

(ii) explain how mutant *bcr/abl* gene contributes to cancer development in CML. [2]

1. *bcr/abl* gene expresses a mutant tyrosine kinase with increased activity and
2. is continuously activated which leads to unregulated cell division.
3. cells with *bcr/abl* fusion protein are also resistant to apoptosis induced by DNA damage, hence allowing accumulation of mutations that lead to cancer.

(e) Suggest a therapeutic strategy that can be used to treat CML caused by Philadelphia chromosome. [1]

1. Use of Tyrosine kinase inhibitor to reduce increased tyrosine kinase activity of the mutant *bcr/abl* protein.

[Total: 10]

- 6 In the inheritance of feather colour in chickens, Individuals carrying the dominant allele, **I**, have white plumage even if they also carry the dominant allele, **C**, for coloured plumage.

White Leghorn chickens have the genotype **IIcc**, while white Wyandotte chickens have the genotype **iiCC**.

A white Leghorn is crossed with a white Wyandotte and the F_2 generation yield 27 chickens with white plumage and 7 chickens with coloured plumage.

- (a) Use a genetic diagram to show the expected genotypes and phenotypic ratios in the F_2 generation. [4]

F₁ phenotype: White Chicken x White Chicken

F₁ genotype (2n): IiCc x IiCc

Gametes (n): IC, Ic, iC, ic x IC, Ic, iC, ic

F₂ genotype and phenotype (2n):

♀ \ ♂	IC	Ic	iC	ic
IC	IIcc White	IiCc White	IiCC White	IiCc White
Ic	IIcc White	Iicc White	IiCc White	Iicc White
iC	IiCC White	IiCc White	iiCC Coloured	iiCc Coloured
ic	IiCc White	Iicc White	iiCc Coloured	iiCC White

F₂ phenotypic ratio: 13 White Plumage : 3 Coloured Plumage

- (b) (i) State the name for this type of interaction between gene loci. [1]

- Dominant suppression epistasis / Epistasis

- (ii) Explain why gene locus C cannot be expressed in the presence of dominant allele I. [1]

- Gene product of the dominant I allele inhibits the phenotypic effect of the C allele

- (c) Explain how you could determine that a chicken with coloured plumage was homozygous at both loci. [3]

1. A chicken with white plumage and homozygous recessive at both loci **iiCC** is used to cross with this chicken with coloured plumage but unknown genotype.
2. If all of the offspring have coloured plumage, the genotype of the chicken is **iiCC** / homozygous at both loci.
3. If some of the offspring have white plumage, the genotype of the chicken is **iiCc** / not homozygous at both loci.

(d) Given that the calculated chi-square value for this cross is 0.098 and the chi-square table given below. Conclude if the observed results follow the expected phenotypic ratio at 5% level of significance. [1] L2

Chi-square table

Degrees of Freedom (df)	Probability (p)										
	0.95	0.90	0.80	0.70	0.50	0.30	0.20	0.10	0.05	0.01	0.001
1	0.004	0.02	0.06	0.15	0.46	1.07	1.64	2.71	3.84	6.64	10.83
2	0.10	0.21	0.45	0.71	1.39	2.41	3.22	4.60	5.99	9.21	13.82
3	0.35	0.58	1.01	1.42	2.37	3.66	4.64	6.25	7.82	11.34	16.27
4	0.71	1.06	1.65	2.20	3.36	4.88	5.99	7.78	9.49	13.28	18.47

- Since $\chi^2_{\text{calculated}} (= 0.098) < \chi^2_{\text{critical}} (= 3.84)$ at $p=0.05$ and $v = 1$, there is no significant difference between observed and expected ratio of 13 white plumage: 3 coloured plumage / any difference between observed and expected ratio of 13 white plumage: 3 coloured plumage is due to chance

Or

- Since the p value of $\chi^2_{\text{calculated}} > p= 0.05$ of χ^2_{critical} , there is no significant difference between observed and expected ratio of 13 white plumage: 3 coloured plumage / any difference between observed and expected ratio of 13 white plumage: 3 coloured plumage is due to chance

[Total: 10]

- 7 Parkinson's disease is a neurodegenerative disorder which leads to progressive deterioration of motor function due to loss of dopamine-producing brain cells.

A substance called dopamine acts as a messenger between two brain areas - the substantia nigra and the corpus striatum - to produce smooth, controlled movements.

Most of the movement-related symptoms of Parkinson's disease are caused by a lack of dopamine due to the loss of dopamine-producing cells in the substantia nigra.

When the amount of dopamine is too low, communication between the substantia nigra and corpus striatum becomes ineffective, and movement becomes impaired; the greater the loss of dopamine, the worse the movement-related symptoms.

Other cells in the brain also degenerate to some degree and may contribute to non-movement related symptoms of Parkinson's disease.

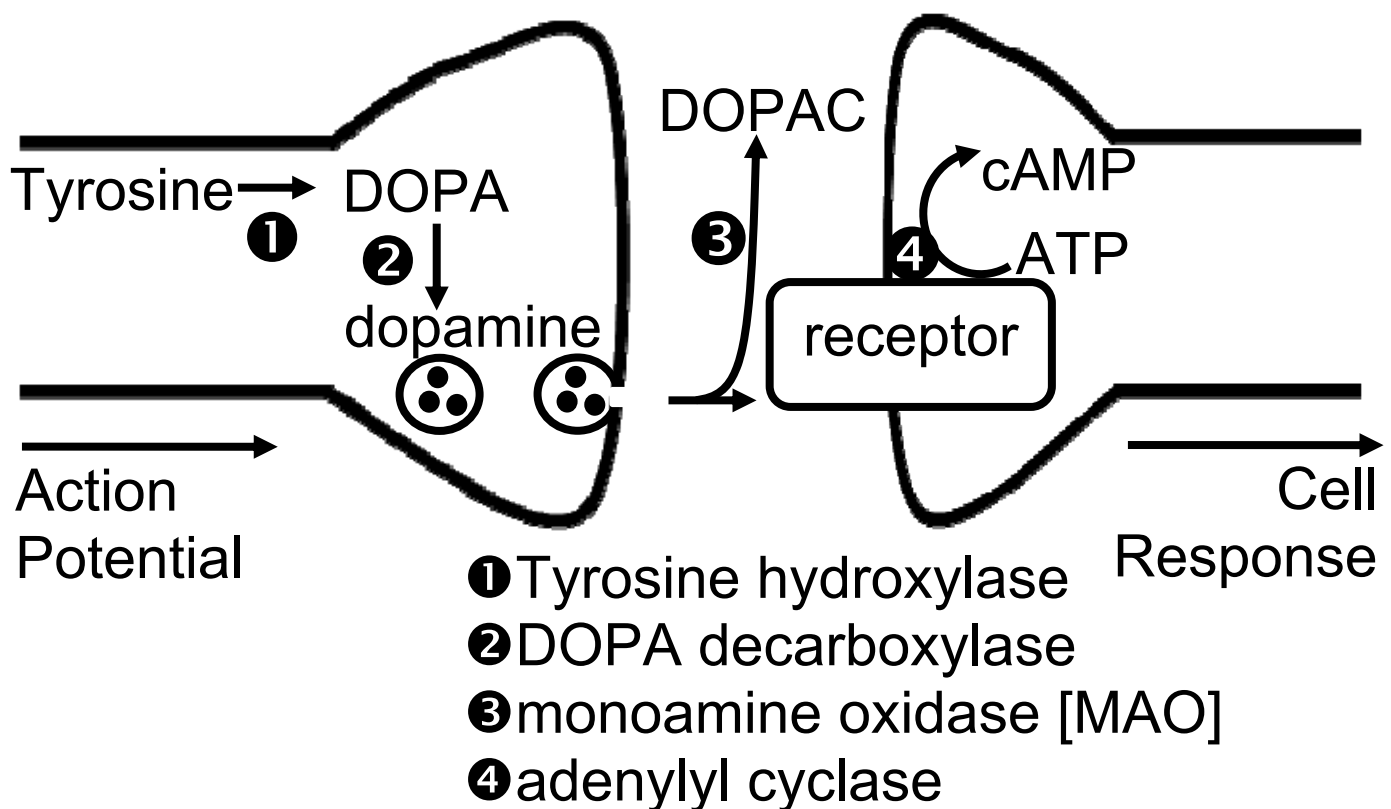


Fig 7.1

(a) Explain the role of dopamine in normal motor functioning

[2]

1. Dopamine is a neurotransmitter in nerve cells released from nerve cells in substantia nigra that binds to complementary receptors on post synaptic membrane in corpus striatum.

2. Dopamine is important in allowing transmission of nervous information to maintain effective communication between substantia nigra and corpus striatum that coordinates smooth motor function / controlled movements

(b) With reference to Fig. 7.1 suggest **three** theoretical treatments of Parkinson's disease with the use of 'designer' enhancing or inhibiting drugs [3]

1. aim / principal is the maintenance of dopamine in the synapse so the level of communication between the two brain areas - the substantia nigra and the corpus striatum can be regulated to produce smooth, controlled movements- By inhibiting effects of enzyme MAO so that less dopamine is broken down.
2. to produce drugs structurally similar to / complement / augment the effects of dopamine.
3. Up-regulate the gene expression of enzymes Tyrosine hydroxylase and DOPA decarboxylase in remaining dopaminergic nerves to increase dopamine production levels / production levels and communication across the synapse is maintained.

Fig. 7.2 and 7.3 compare the cells within the substantia nigra in patients with Parkinsons disease and normal individuals.

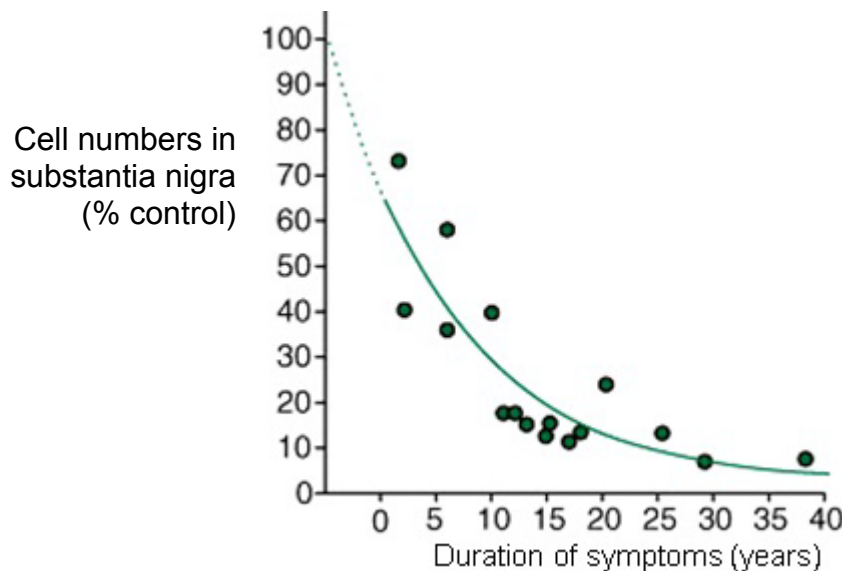


Fig 7.2 Cell numbers in substantia nigra in patients with Parkinson Disease compared over

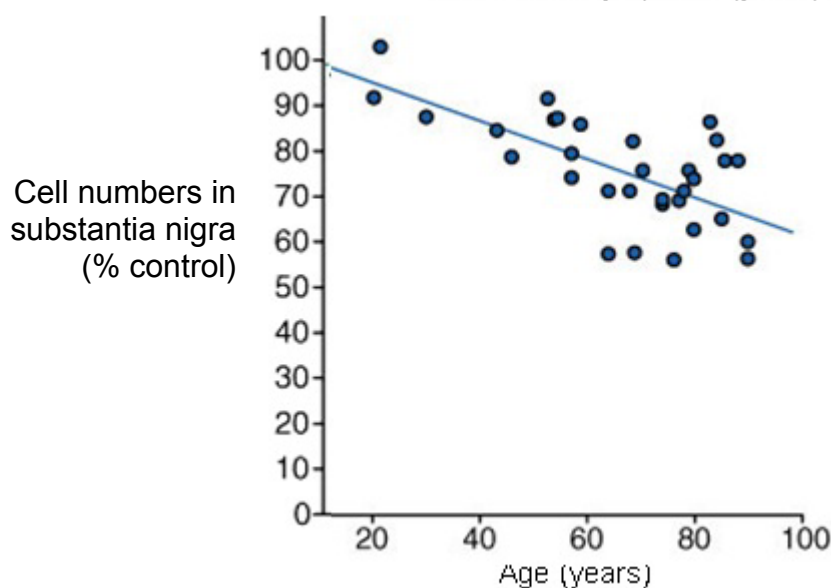


Fig 7.3 Cell numbers in Substantia nigra in normal individuals compared over time

(c) With reference to figures 7.1, 7.2 and 7.3 explain the factors leading to Parkinson's disease [3]

1. In patients with Parkinson's Disease [PD] there is a 70% (70% over 40 years) decline in cell numbers in the substantia nigra compared to 4% / (750-650 over 40 years) decline in normal persons.
2. this would decrease the number of cells producing dopamine, resulting in a low concentration of Dopamine.
3. dopamine is important in communication between the substantia nigra and the corpus striatum for smooth motor control.
4. with the decline in dopamine levels there is a *decrease in the communication between the substantia nigra and the corpus striatum which leads to an increase in PD symptoms.*

(d) With reference to Fig 7.2,
Suggest why upon the onset of Parkinson's disease at year '0' the cell count in substantia nigra is at 70%. [2]

1. PD involves the gradual loss of dopamine producing nerve cells
2. effects of PD are not experienced till a critical threshold of substantia nigra nerve cells has been crossed / OWTTE
3. at time of sampling [year '0'], patients are already display the symptoms of PD which means that they already have crossed the threshold of substantia nigra cell death ie 70% .

[Total: 10]

8 (a) With reference to Fig 8.1 and 8.2, complete the taxonomy of 'Neopro' : [1]

Genus : _____

Species : _____

GENUS : Neolamprologus [Reject if not underlined and if misspelt]

SPECIES : prochilus [Reject if not underlined and if misspelt]

(b) Explain the possible evolutionary processes that could have led to the many species seen in Fig 8.2. [3]

1. Evolution of many species via Adaptive radiation over a relatively short period of time.
2. due to different selection pressures of food sources, habitat etc
3. Sympatric speciation- no geographical isolation.
4. prezygotic barriers- behavioural / temporal
5. over time become reproductively isolated

(c) Explain how DNA homology is used to predict phylogeny. [2]

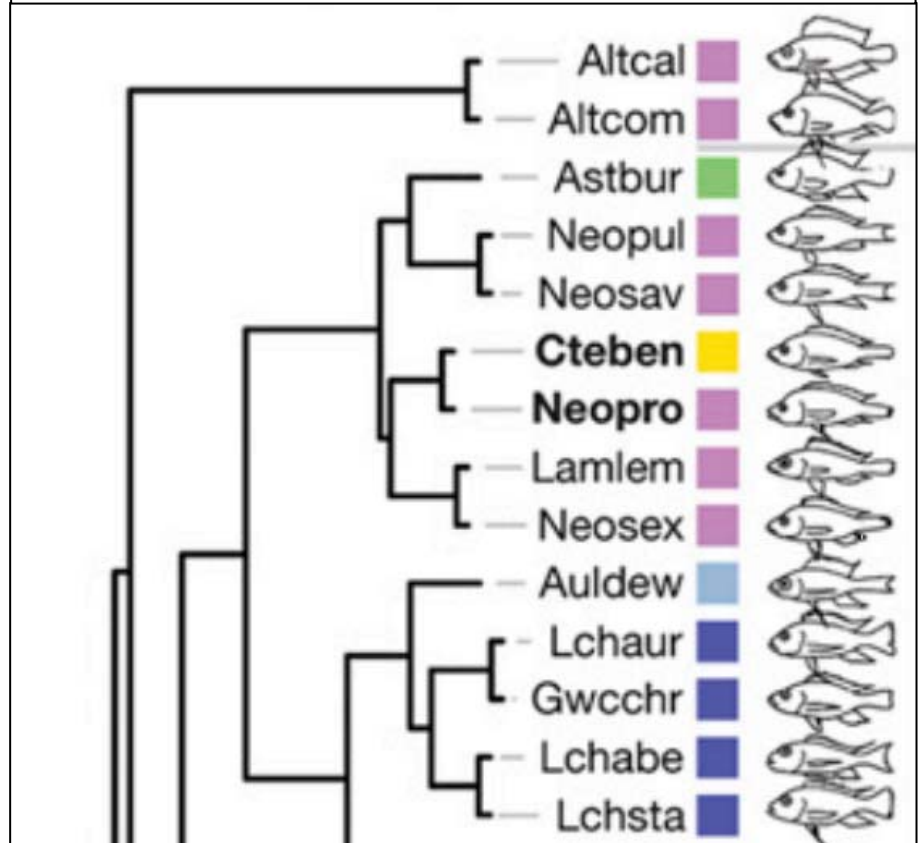
1. DNA homology is the study of the degree of similarity between the two types of species being studied.
2. DNA homology is useful as most organisms (less Viruses) possess DNA and therefore is a good level for comparison.
3. the principle for comparison- the higher the homology, the greater the number of complementary nucleotide base pairs, the higher the temperature needed to separate the two strands of DNA. the higher the homology the more related two species are.
4. the phylogenetic relationship between them can further be determined with the greater degree of similarity.

(A) body plan / shape

Fig 8.3

(A) Portion of the Cluster analysis * proposing phylogenetic relationships on the basis of 17 landmarks on body shape. Outlines in (A) are based on real photographs

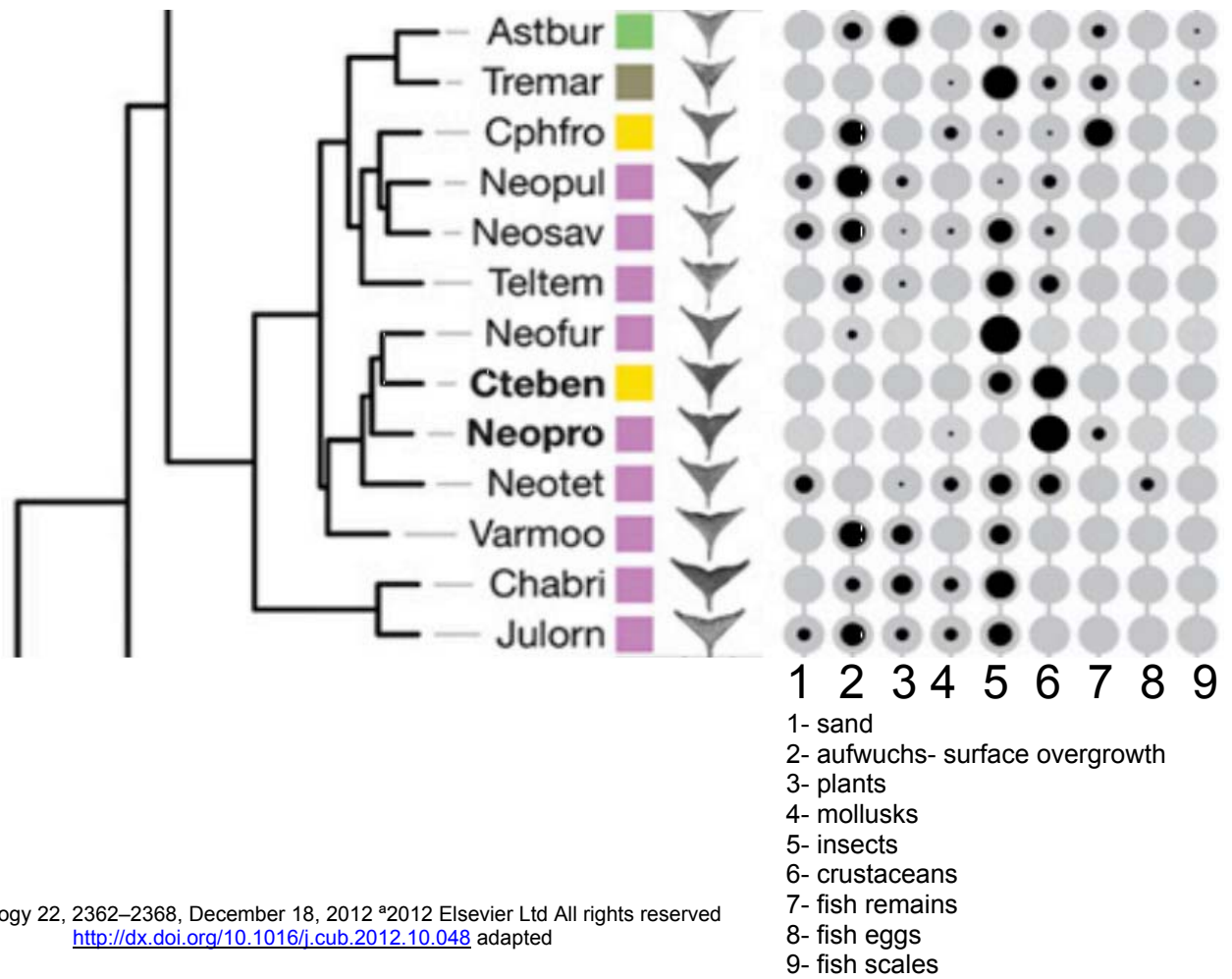
*Cluster Analysis is a systematic comparison of parameters that allows a conclusion on relationships between specimens.



Current Biology 22, 2362–2368, December 18, 2012 ©2012 Elsevier Ltd All rights reserved <http://dx.doi.org/10.1016/j.cub.2012.10.048> adapted

(B) lower pharyngeal jaw bone

(C) gut content



Current Biology 22, 2362–2368, December 18, 2012 ©2012 Elsevier Ltd All rights reserved
<http://dx.doi.org/10.1016/j.cub.2012.10.048> adapted

Fig. 8.4

(B) Portion of the Cluster analysis proposing phylogenetic relationships on the basis of landmarks on the lower pharyngeal jaw bone. images in (B) are taken from dissected LPJs
(C) Results from the stomach and gut content analyses (in volume %).

(d) Explain the significance of the results found in Fig 8.2, 8.3 and 8.4 on the phylogeny of 'Neopro' and 'Cteben' species. [4]

1. in Fig 8.2 comparing **homology of DNA** place Neopro and Ctebens as **separate Genera**.
2. in Fig 8.3 comparing **homology of Body plans** place Neopro and Ctebens as more closely related arising from the **same speciation event / i.e. same genus**
3. in Fig 8.4 comparing **homology of jaw bones** place Neopro as a **common ancestor** to Ctebens / i.e. **same genus**.
4. Results in **Fig 8.3 and Fig 8.4** suggest that Neopro and Ctebens are more **closely related** whereas **Fig 8.2** suggest that Neopro and Ctebens are more **distantly related**
5. Possible **convergent evolution**
6. possibly due to the **same selection pressures of food sources- crustaceans**.

[Total: 10]

Section B

Answer **all** questions. Answer each part on a **separate** piece of paper.

Write your answers on separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

9 (a) Compare the lytic and lysogenic cycles of bacteriophages.

[6]

Basis	Lytic	Lysogenic
• Adsorption	1. Phage adsorbs/attaches to specific receptor sites on the bacterial cell wall.	
• Penetration	2. Phage injects the viral genome into the bacterium's cytoplasm.	
• Replication	3. Both requires utilization of host cell machinery for transcription, translation or DNA replication.	
• Replication	4. Phage directs the host cell to replicate the phage DNA and synthesise phage proteins. No prophage resulted from this cycle.	4. Phage intergrates its viral genome into a specific site in the host genome to become a prophage.
• Replication	5. Many viral genes are expressed to synthesise phage enzymes and structural components for assembly of new viruses.	5. One prophage gene which codes for a repressor protein that prevents transcription of the other viral genes is expressed.
• Maturation & Release	6. Newly-assembled phages formed are ready and will be released by the host cell.	6. No newly formed phages are formed. Instead, phage DNA replicates as part of bacterial DNA (in the form of prophage).
• Death of host cell	7. Phage codes for enzymes that degrade the host's DNA and lyses the cell.	7. Phage DNA is replicated together with the host's DNA without killing the cell.
• Death of host cell	8. Host cell dies upon release of new viruses.	8. Host cell survives and every daughter bacterium now contains a prophage.
• Switching of cycles	9. Viruses undergoing lytic cycle cannot switch to lysogenic.	9. Viruses undergoing lysogenic cycle can switch to lytic.

(b) Describe the reproductive cycle of the influenza virus.

[8]

1. The influenza virus is acquired by inhalation or ingestion of virus-contaminated respiratory secretions
2. Haemagglutinin glycoprotein on the envelope of the influenza virus binds to specific receptors [sialic acid] on the host's cell surface membrane.
3. After attachment, the cell's cell surface membrane bulge inward, seal off and the virus enters cell via a vesicle through receptor-mediated endocytosis.
4. The virus is now enclosed in an endocytic vesicle. Fusion occurs between the membrane of the endocytic vesicle and viral envelope. The capsid is released into the cytosol.
5. Cellular enzymes removes the capsid. The virus then dispatches its genetic material (made up of RNA) and its internal proteins to the nucleus of the cell.
6. The viral genome functions as a template for making complementary RNA strands via its RNA-dependent RNA polymerases.
7. Complementary RNA strands serve as templates for making new copies of genome RNA and serve as mRNA for production of viral proteins.
8. mRNA is translated on the ribosomes of rough endoplasmic reticulum into capsid proteins, enzymes and glycoproteins for the viral envelope.
9. Vesicles transport the HA glycoproteins and the NA enzyme to the cell's cell surface membrane.
10. After the capsid and viral genome self-assemble, they bud from the host cell covered with an envelope derived from the host's cell surface membrane which has viral glycoproteins inserted.
11. Sialic acid is removed from the envelopes of new viral particles through neuraminidase and this facilitates the release of viral particles from host cell.

(c) Distinguish prokaryotic and eukaryotic gene structure & regulation.

[6]

Basis	Prokaryotic	Eukaryotic
• Absence/presence of operons	1. Genes involved in the same metabolic reaction are grouped together in an operon system that typically comprises a promoter, operator and a regulatory gene.	1. Genes are not grouped together in an operon but are instead occur singly and they require regulatory sequences such as promoters, enhancers & silencers.
• Absence/presence of introns	2. Prokaryotic genes do not or rarely contain introns.	2. Eukaryotic genes usually contain introns.
• Promoter	3. The prokaryotic promoter contains a Pribnow box in which the RNA polymerase will bind to.	3. The eukaryotic promoter contains a TATA box in which general transcription factors will bind to.
• Monocistronic vs Polycistronic	4. Most prokaryotic genes generate polycistronic mRNA after transcription whereby genes which are involved in the same metabolic reaction are grouped	4. Genes generate monocistronic mRNA after transcription in which each gene is regulated by a distinct promoter.

	together and regulated under the control of one promoter.	
• Location	5. Gene regulation takes place in the nucleoid.	5. Gene regulation takes place in the nucleus.
• Coupling of translation to transcription	6. Transcription & translation can occur at the same time and location.	6. Genes have to be transcribed first in the nucleus before translation can take place in the cytoplasm.
• Gene repression	7. Structural genes can be repressed when active repressor binds to the operator which blocks RNA polymerase from binding to the promoter.	7. Genes can be repressed when the transcriptional repressor protein binds to the silencer which may block the binding of transcriptional activator protein from binding to the enhancer.
• Enhancement of gene transcription	8. Genes transcription may be enhanced when cAMP-CAP complex binds to the CAP-binding site (e.g. from lac operon), which increases affinity of RNA polymerase to the promoter.	8. Gene transcription may be enhanced when transcriptional activator proteins bind to the enhancer & DNA bending will occur followed by mediator proteins bringing the complexes formed at the enhancer and promoter together.

10(a) Describe how carbohydrates are formed in the Calvin cycle.

[7]

1. CO₂ combines with 5C sugar ribulose biphosphate (RuBP);
2. Carbon fixation catalysed by RuBP carboxylase/Rubisco;
3. 6C intermediate splits to form two glycerate-3-phosphate (GP);
4. use of ATP (in carbon reduction);
5. reduction / use of NADPH (in carbon reduction);
6. to form 1 glyceraldehyde-3-phosphate (GALP/triose phosphate);
7. GALP/G3P used to regenerate RuBP,
8. utilising ATP molecules (in RuBP regeneration);
9. GALP/G3P/triose phosphate used for synthesis of carbohydrate;

10(b) Outline the role of NAD and NADP in cells.

[7]

1. NADP/ NAD as electron/hydrogen carrier or coenzymes/ carry hydrogen atoms from one compound to another;

Aerobic respiration (NAD):

2. Glycolysis: Accept H⁺ from triose phosphate/ glyceraldehyde phosphate, which is oxidized to 1,3 diphosphoglycerate or pyruvate;
3. Forms 2 NADH for each glucose molecule;
4. Link reaction: Accept H⁺ from pyruvate, which is converted to acetyl CoA;
5. Forms 2 NADH for each glucose molecule;
6. Krebs cycle: Accept H⁺ from intermediates in the regeneration of oxaloacetate from citrate;
7. Forms 6 NADH for each glucose molecule;
8. Link rxn and Krebs cycle occurs in the mitochondrial matrix;
9. Carry hydrogen/ electrons to ETC;
10. Oxidative phosphorylation/ transfer of electrons down the ETC to release ATP;
11. 1 NADH yields 3/2.5 ATP;
12. NAD reoxidised in oxidative phosphorylation;

Anaerobic respiration (NAD):

13. NAD regenerated when electrons from NADH transferred to pyruvate/ ethanal;
14. NAD reused in glycolysis to generate 2 ATP;

NADP:

15. Final electron acceptor in non-cyclic photophosphorylation;
16. Drives photolysis of water;
17. $2\text{H}^+ + 2\text{e}^- + \text{NADP} \rightarrow \text{NADPH}$ or reduced NADP/ hydrogen ions from splitting of water
18. combine with NADP to form NADPH;
19. NADPH passes to Calvin cycle/ light independent reactions in the stroma;
20. NADPH reduces glycerate biphosphate/1,3 diphosphoglycerate to triose phosphate/
21. glyceraldehyde phosphate and gets reoxidized back to NADP;

10(c) Compare and contrast the process of electron transport through the electron transport chain in the mitochondrion and chloroplast. [6]

Basis of comparison	
Similarities (basis of comparison) – [Max 3]:	- Series of electron carriers / cytochrome are of decreasing energy levels; - Energy released is used to actively pump H^+ across membrane; - to create a H^+ gradient / proton motive force; - Which is used to drive ATP synthesis as H^+ return via ATP synthase down the H^+ gradient / chemiosmosis; - Both ETCs are found on membranes of organelles. [Max 3]
Differences (basis of comparison) – [Max 3]:	ETC in Mitochondrion
6. Transfer of electrons to ETC	From reaction centre / chlorophyll a / / electron acceptor of photosystems / primary pigment
7. Final electron acceptor	NADP ⁺
8. Products formed	H ₂ O, ATP, NAD ⁺ and FAD ⁺
9. Location	Thylakoid membrane
10.Direction of proton pumping	From stroma into thylakoid lumen / thylakoid space
11.Direction of proton movement down concentration gradient	From thylakoid lumen into stroma
12.Nature of ETC	Electron transport is non-cyclic
13.Source of energy	Light energy
14.Nature of metabolic process	Involved in anabolic process
15.No. of ETC	Two in non-cyclic photophosphorylation

END OF PAPER